

## Akshera Paladhi

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## EDUCATION

University of California Santa Cruz, Santa Cruz, CA

Bachelor of Science in Electrical Engineering B.S., Minor in Astrophysics

September 2024 – Present

Expected Graduation June 2028

GPA: 3.73/4.0

- **Relevant Coursework:** Computer Systems and Assembly Language, Modern Electronics, Introduction to Physics I, Computer Systems and C Programming, Linear Algebra, Vector Calculus, Python Programming, Introduction to Probability Theory, Physics III Introduction to Electricity and Magnetism, Introduction to Scientific Computing

## EXPERIENCE

Air Mouse (ESP32 + MPU-6050), (Remote)

December 2025 – Present

Arduino IDE/PCB Project

- Developed a motion-controlled wireless mouse using ESP32 and MPU-6050, implementing **BLE HID**, **gyro calibration**, **moving average filtering**, and **adjustable sensitivity/deadzone for stable, real-time cursor control**
- Built and testing prototype on **Arduino/breadboard**; currently **developing PCB**, demonstrating **embedded programming** and **sensor integration skills**.

Learning Support Services Tutor (Hybrid)

Sept 2025 – Present

Large Group Tutor/Individual Tutor

- Leading weekly tutoring sessions for 5+ students in CSE 12, reinforcing **low-level logic**, **data structures**, **RISC-V architecture**, and system-level understanding to improve retention and applied learning
- Providing one-on-one support in **Linear Algebra**, strengthening conceptual reasoning in **vector spaces**, **eigenvalues**, and **matrix computation**

UC Santa Cruz Quantum Computing (Hybrid)

August 2025 – Present

Vice President/Researcher

- Organizing weekly technical workshops and events for 20+ student members, introducing practical applications of **quantum algorithms**
- Collaborating with leadership to expand research opportunities, overseeing a project comparing quantum neural networks (**QNNs**) with deep neural networks (**DNNs**) for **data-driven analysis**

UC Santa Cruz Institute of Particle Physics (SCIPP) (Hybrid)

June 2025 – Present

Undergraduate Researcher

- Developing **embedded C/C++** firmware for **STM32** with **UART/I<sup>2</sup>C** for **real-time data acquisition**
- Debugging **hardware and software systems** using **oscilloscopes**, **logic analyzers**, and **multimeters**
- Automating **calibration** and logging in Python, increasing hardware testing throughout
- Contributing to a **Git-managed codebase**, following professional collaboration practices

L'Space Nasa's Proposal Writing and Evaluation Experience (Remote)

May 2025 – August 2025

Research Participant (Workforce Preparation)

- Served as Scientist on a student team in NASA's Workforce Preparation Academy, developing science goals and traceability matrices, conducting **trade studies**, and creating **CAD designs** and using **Siemens NX**
- Reviewed **NASA solicitation proposals**, evaluating submissions against rubric criteria and providing structured feedback to proposal teams

SSCS Arduino Contest - IEEE Project, (Hybrid)

April 2025 – August 2025

Research Project

- Designed and implemented **embedded C/C++** for a wearable audio-to-haptic device, ensuring real-time performance with analog filtering and **UART/I<sup>2</sup>C** communication
- Led **hardware-software integration** and testing to validate system accuracy and robustness for accessibility applications
- Applied circuit design and **motor control** principles to optimize power and system reliability

## SKILLS

- **Programming & Software:** C, C++, Embedded C, Python, MATLAB, HTML/CSS/JavaScript, Jupyter Notebooks, Git/GitHub, Linux, Firmware Development, Scientific Computing
- **Hardware & Engineering Tools:** STM32 microcontrollers, UART/I<sup>2</sup>C/SPI, KiCad, LTspice, PCB Design, Thorlabs CC215MU camera (ThorImageCAM/ThorCam), Machine Learning, Signal Processing, Mathematical Optimization, Data Structures & Algorithms, Probability, Circuit Analysis, Orbital Mechanics, FFT / Frequency-Domain Analysis